

Feb 12 2026

$$(2) \quad |T(x) - T(y)| < \rho |x - y|, \quad \rho < 1$$

$$\forall x, y \in U(\delta, r)$$

$$|T(x) - T(y)| = \max_{t \in [0, \delta]} |T(x)(t) - T(y)(t)|$$

$$\begin{aligned} |T(x)(t) - T(y)(t)| &= \left| \int_0^t f(\tau, x(\tau)) d\tau - \int_0^t f(\tau, y(\tau)) d\tau \right| \\ &\leq \int_0^t |f(\tau, x(\tau)) - f(\tau, y(\tau))| d\tau \\ &\leq \int_0^t L |x(\tau) - y(\tau)| d\tau \end{aligned}$$

$$\text{Recall } x, y \in U(\delta)$$

$$\Rightarrow |x(\tau) - x_0| \leq r \quad \forall \tau$$

$$|y(\tau) - x_0| \leq r \quad \forall \tau$$

$$\Rightarrow |x(\tau) - y(\tau)| \leq 2r \quad \forall \tau$$

$$|T(x)(t) - T(y)(t)| \leq \int_0^t L |x(\tau) - y(\tau)| d\tau$$

$$|x(\tau) - y(\tau)| \leq |x - y|$$

$$|T(x)(t) - T(y)(t)| \leq L |x - y| t$$

$$\leq \underbrace{L \delta}_{\rho} |x - y|$$

$$\text{we need } \rho < 1 \Rightarrow \delta L < 1$$

$$\Rightarrow \delta < \frac{1}{L}$$

$$\text{Last time } \delta \leq \frac{r}{Lr + h}$$

$$\frac{r}{Lr + h} \leq \frac{1}{L}$$

$$Lr \leq Lr + h \quad \checkmark$$

$$\therefore \boxed{\delta \leq \frac{r}{Lr + h}} \text{ ensures invariance and contraction}$$

Contraction Mapping Theorem:

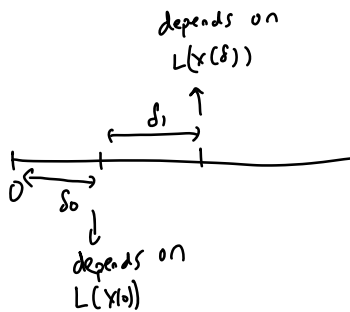
$$\Rightarrow \exists \text{ unique } x \in U(\delta) \text{ which is a fixed point of } x = T(x)$$

"globally attractive" from any $x^{(0)} \in U(\delta)$

$$x^{(0)} = T(x^{(0)})$$

If $f(t, x)$ is continuous in t and Lipschitz in x

$\Rightarrow \dot{x} = f(t, x), x(0) = x_0$ has unique solution in $[0, \delta]$ for $\delta \leq \frac{r}{Lr+h}$



Because δ depends on $L(x_0)$,

we cannot readily claim existence and uniqueness over $(0, \infty)$

Recall

$$\dot{x} = x^2$$

$$f(x) = x^2$$

$$|f(x) - f(y)| = |x^2 - y^2|$$

$$= |x - y| |x + y|$$

If in addition, $f(t, x)$ is globally Lipschitz in x ,

then $\exists$ unique solution for all t .

$$\dot{x} = -x^3$$

$$\frac{dx}{dt} = -x^3$$

$$\frac{-dx}{x^3} = dt$$

$$\frac{x^{-2}}{2} \Big|_{x_0}^x = t$$

$$x^{-2} - x_0^{-2} = 2t$$

$$x^{-2} = 2t + x_0^{-2}$$

$$\therefore x^2 = \frac{1}{2t + x_0^{-2}}$$

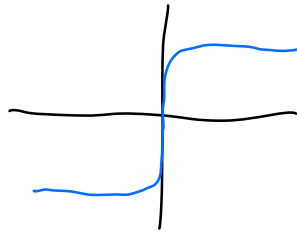
$$x = \text{sgn}(x_0) \sqrt{\frac{1}{2t + \frac{1}{x_0^2}}}$$

$f(x) = x^3$ is not globally Lipschitz and yet a unique solution for all t

Ex. $\dot{x} = x^{1/3}, x(1) = 0$
 $= f(x)$

$$\frac{df}{dx} = \frac{1}{3} x^{-2/3} \Big|_{x=0} = \infty$$

Not Lipschitz



$$x(t) \geq 0$$

$$x(t) = \left(\frac{2t}{3}\right)^{3/2}$$

$$\dot{x} = \frac{3}{2} \left(\frac{2}{3}t\right)^{1/2} \cdot \frac{2}{3}$$

$$= \left(\frac{2}{3}t\right)^{1/2}$$

$$= \left[\left(\frac{2}{3}t\right)^{3/2}\right]^{1/3}$$

$$= x^{1/3}$$

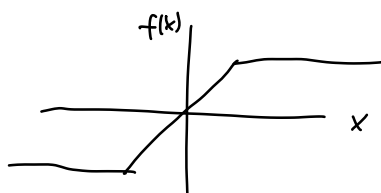
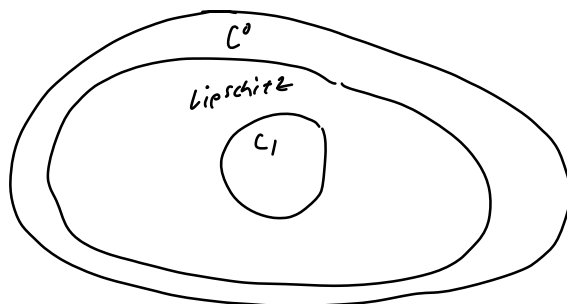
C^0 : continuous
 $\downarrow$

Lipschitz

$\uparrow$

C^1 : continuous differentiable

s.s. previously C^1



$$\dot{x} = f(t, x; \mu) \quad x(0) = x_0$$

Continuous dependence on initial condition:

$$\dot{x} = f(t, x)$$

x_0, y_0 : two initial conditions
 $\downarrow \quad \downarrow$
 $x(t) \quad y(t)$

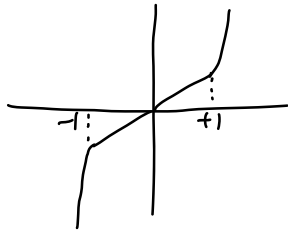
continuous dependence on $[0, \tau]$

for any $\varepsilon > 0$, $\exists \delta(\varepsilon, \tau) > 0$

$$\text{s.t.} \quad |x_0 - y_0| \leq \delta \Rightarrow |x(t) - y(t)| \leq \varepsilon \quad \forall t \in [0, \tau]$$

$\Uparrow$

if $x(t), y(t)$ remain in a set with Lipschitz constant L over $[0, \tau]$



$$\dot{x} = f(t, x; \mu)$$

$$\dot{\mu} = 0$$

$$X = \begin{bmatrix} x \\ \mu \end{bmatrix}$$

$$X_0 = \begin{bmatrix} x_0 \\ \mu \end{bmatrix}$$

$$\dot{X} = F(t, X)$$

$$F = \begin{bmatrix} f(t, X) \\ 0 \end{bmatrix}$$

Sensitivity to parameters

$$\dot{x} = f(t, x, \mu) \quad x(0) = x_0$$

$$\mu = \mu_0 \text{ (nominal)}$$

$\phi(t, x_0; \mu) : x(t)$ starting from initial condition x_0 with model parameters μ

$$x_0 + \int_0^t f(t, x(\tau); \mu) d\tau$$

sensitivity matrix:

$$(n \times p) \rightarrow S(t, x_0, \mu) := \frac{\partial \phi(t, x_0, \mu)}{\partial \mu} \Big|_{t=0}$$

$$x \in \mathbb{R}^n \rightarrow \phi \in \mathbb{R}^n$$

$$\mu \in \mathbb{R}^p$$

$$S = \begin{bmatrix} \frac{\partial \phi_1}{\partial \mu_1} & \dots & \frac{\partial \phi_1}{\partial \mu_p} \\ \dots & \dots & \dots \\ \frac{\partial \phi_n}{\partial \mu_1} & \dots & \frac{\partial \phi_n}{\partial \mu_p} \end{bmatrix}$$

$$\phi(t, x_0, \mu) = \begin{bmatrix} \phi_1(t, x_0, \mu) \\ \dots \\ \phi_n(t, x_0, \mu) \end{bmatrix} \leftarrow \begin{matrix} x_1 \\ \dots \\ x_n \end{matrix}$$

$$\phi(t, x_0, \mu) = x_0 + \int_0^t f(\tau, \phi(\tau), \mu) d\tau$$

$\Downarrow$

$$\frac{\partial \phi}{\partial t} = f(t, \phi(t), \mu)$$

$$\frac{\partial S}{\partial t} = \frac{\partial}{\partial t} \frac{\partial \phi}{\partial \mu}$$

$$= \frac{d}{d\mu} \frac{d\phi}{dt}$$

$$= \frac{d}{d\mu} f(t, \phi(t), \mu)$$

$$= \frac{d}{d\mu} f(t, \phi(t, x_0, \mu), \mu)$$

$$= \frac{\partial}{\partial \mu} f(t, \phi(t, \mu), \mu) + \frac{\partial f(t, \phi, \mu)}{\partial \phi} \frac{\partial \phi}{\partial \mu}$$

$S(0, x_0, \mu)$: initial condition